THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE UNIT TEST SET B

Chapters 1–4: **Exploring Substances: Acidic, Basic, and Neutral, Electricity: Circuits and Their Components, The World of Metals and Non-metals, Changes Around Us: Physical and Chemical** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–12 are Objective Type (1 mark each).
3. Q. 13–17 are Very Short Answer Type (2 marks each).
4. Q. 18–23 are Short Answer Type (3 marks each).
5. Q. 24–25 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 22 to 25.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 12 × 1 = 12

Choose the correct option and write it in your answer book.

1. Turmeric paper turns ___ with a basic substance. 1

(a) colourless **(b)** reddish-brown **(c)** blue **(d)** green

2. To neutralise acidic soil, farmers add: 1

(a) vinegar **(b)** sugar

(c) lime (slaked lime) **(d)** common salt

3. A solution turns blue litmus red and does not change red litmus. It is: 1

(a) acidic **(b)** basic **(c)** neutral **(d)** a salt

4. A bulb connected to a cell glows because: 1

(a) current heats the filament **(b)** the switch emits light

(c) the cell shines **(d)** wires glow

5. Which of the following is an insulator? 1

(a) Aluminium foil **(b)** Plastic **(c)** Graphite **(d)** Iron nail

6. The current from a cell is: 1

(a) high voltage AC **(b)** mains current

(c) direct current (DC) **(d)** alternating current

7. The non-metal that is the main constituent of coal is: 1

(a) carbon **(b)** iodine **(c)** sulfur **(d)** oxygen

8. The metal present in the red blood pigment is: 1

(a) zinc **(b)** sodium **(c)** iron **(d)** copper

9. An alloy is a: 1

(a) compound of gases

(b) mixture of metals (or a metal and non-metal)

(c) non-metal only

(d) pure metal

10. The change in which a metal gets a green coating is: 1

(a) melting **(b)** freezing

(c) dissolving **(d)** corrosion (chemical change)

11. Changes which are good for us, e.g. ripening fruits, are: 1

(a) slow only **(b)** harmful **(c)** desirable **(d)** undesirable

12. Boiling water to make steam is: 1

(a) a physical change **(b)** combustion
(c) a chemical change **(d)** rusting

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 15–25 words each.

13. Why is baking soda applied on an ant bite? 2

14. What is a filament? What happens when current flows through it? 2

15. What is sonority? Give an example. 2

16. Define malleability and ductility with examples. 2

17. Name two examples of chemical changes in the kitchen. 2

SECTION — C (Short Answer Type) 6 × 3 = 18

Answer in about 40–60 words each.

18. What is neutralisation? Describe the experiment with lemon juice and lime water. 3

19. Describe how to make a simple electric circuit with a cell, bulb, wire and switch. 3

20. A metal strip is hit with a hammer. One piece turns to powder while another flattens. Identify each. 3

21. What are the differences between physical and chemical changes (three points)? 3

22. Differentiate between acids and bases on the basis of taste, touch and litmus. 3

OR

Describe an activity to classify lemon juice, soap solution and sugar solution using litmus. 3

23. Why does a torch bulb not glow if the cell is inserted wrongly? 3

OR

How can you test whether a material is a conductor? Describe the activity. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 80–120 words each.

24. (i) Name acids found in five foods. 5
(ii) State their taste.
(iii) Why is it not safe to taste them all?

OR

(i) Define an indicator. 5
(ii) Describe how you would prepare turmeric paper.
(iii) What colours does it show with an acid and a base?

25. **Case study:** A jeweller notes that silver turns black but gold stays bright. 5
(i) Why?
(ii) What is tarnishing?
(iii) Remedy?
(iv) Why are gold and platinum used in jewellery?

OR

(i) Compare metals and non-metals in a table. 5
(ii) Name three exceptions.
(iii) Why do exceptions matter?

END OF QUESTION PAPER Total: 50 Marks